MODEL CARD · NLP CA3 · BUILD · AUDIT · PITCH

Product Review Sentiment Classifier

TF-IDF + Logistic Regression for Women's E-Commerce Clothing Reviews — baseline and contrast-aware variants

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91.5%
Accuracy
identical for both variants

0.95
F1 (positive class)
precision 0.98 · recall 0.92

85.2% → 85.7%
Negative-review recall
baseline → contrast-aware · 95% CI ≈ 82%–88%

0.961
ROC AUC
avg. precision 0.994

Model details

SCIKIT-LEARN

Task	Binary sentiment classification of review text (positive / negative)
Features	TF-IDF on 1–2-grams, English stop-words removed, min_df 5, max_df 0.9, up to 20k features (14,944 kept)
Classifier	Logistic Regression, class_weight = balanced, max_iter 1000, seed 42
Baseline	Cleaned text (lower-case, punctuation and digits stripped)
Contrast-aware	Same pipeline, but the clause after the first contrast word (but, however, although, though, unfortunately, except, yet) is duplicated before vectorising, ~2× weight
Artifacts	baseline_model.joblib · contrast_aware_model.joblib

Test-set performance

N = 3,964

METRIC	BASELINE	CONTRAST-AWARE
Accuracy	0.915	**0.915**
Precision (positive)	0.979	**0.979**
Recall (positive)	0.923	**0.923**
F1 (positive)	0.950	**0.950**
Recall (negative)	**0.852**	**0.857**
Precision (negative)	0.602	**0.601**
ROC AUC	0.961	**0.961**

Accuracy hides the imbalance (88 / 12). Contrast-aware catches 2 more of 474 negative reviews (406 vs 404) — a small gain, not a fix.

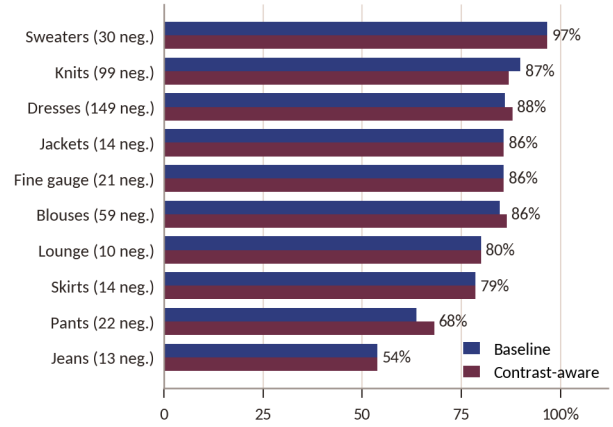
Training & evaluation data

NICK BROOKS · PUBLIC

- **Women's E-Commerce Clothing Reviews:** 23,486 real reviews → **19,818** after dropping empty text and 3-star reviews.
- **Label rule:** ≥ 4 = positive (88.0%), ≤ 2 = negative (12.0%). Median review: 58 words.
- **Split:** stratified 80/20 → 15,854 train / 3,964 test (474 negative, 3,490 positive). One split, no cross-validation.
- **Audit slices:** product category (18) and review length (short ≤ median vs detailed).

Audit 1 · negative recall by category

TEST SET · ≥100 REVIEWS



Labels show each category's negative-review count. Jeans 54% is 7 of 13 (95% CI ≈ 29%–77%), so treat the gap as a flag, not a measurement.

Intended use

IN SCOPE

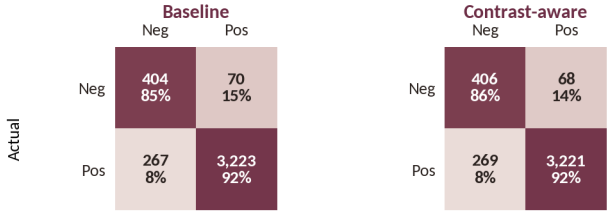
- First-pass triage that surfaces likely complaints in English clothing-review text for a human to read.
- Aggregate sentiment analytics over large review corpora.

OUT OF SCOPE

- Automated action (removal, seller penalties, refunds) without human review.
- Neutral / 3-star sentiment, non-English text, sarcasm, or products outside women's clothing.
- Unreviewed use on Jeans and Pants, where complaint recall is weakest.

Confusion matrices

ROW % = SHARE OF ACTUAL CLASS



Audit 2 · short vs detailed reviews

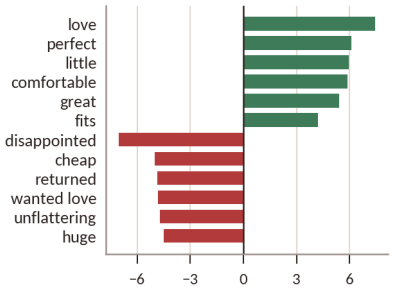
TEST SET

	ACCURACY		NEG. RECALL	
	Base	Contr.	Base	Contr.
Short (n = 2,053)	92.4%	**92.6%**	85.6%	**87.3%**
Detailed (n = 1,911)	90.6%	**90.3%**	84.9%	**84.0%**

All gaps are ≤ 3.3 pts: review length is not where the bias lives; category is.

What drives predictions

CONTRAST-AWARE



Top TF-IDF weights are real sentiment words. **"returned", "huge", "return"** lean negative, hinting at learned sizing / return language that links back to the category gap.

Limitations & risks

- **Category gap:** Jeans 54%, Pants 64→68% vs Sweaters 97%; from only 13 and 22 negative test reviews.
- **Modest fix:** contrast-aware gains under 1 pt overall (2 of 474 reviews); Pants and Dresses improve, Knits drops 3 pts.
- **Precision on negatives is 0.60:** about 4 in 10 flagged reviews are actually positive.
- **Length:** short vs detailed differ by ≤ 3.3 pts, so it is not the main bias.
- **Proxy labels** from star ratings; one dataset, English only, no neutral class, bag-of-words (no negation scope or sarcasm). Reviewer demographics not audited.

Recommendations

- Deploy only as a **first-pass filter with human review**.
- **Tune the threshold toward recall** for Jeans and Pants; track negative recall per category, not just accuracy.
- Re-test the category gap on more negative Jeans / Pants reviews.
- Next: clause-level parsing, negation-scope detection, BERT.

Mixed-signal reviews (praise + "but"): 44.8% of reviews, 40.3% of negatives. Negative recall 82.6% → 84.2%. VADER: 19%. *(project deck)*